

AI/ML Internship Final Project

AI-Based Virtual Try-On and Fashion Recommendation System

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Raritone Summer Internship 2026

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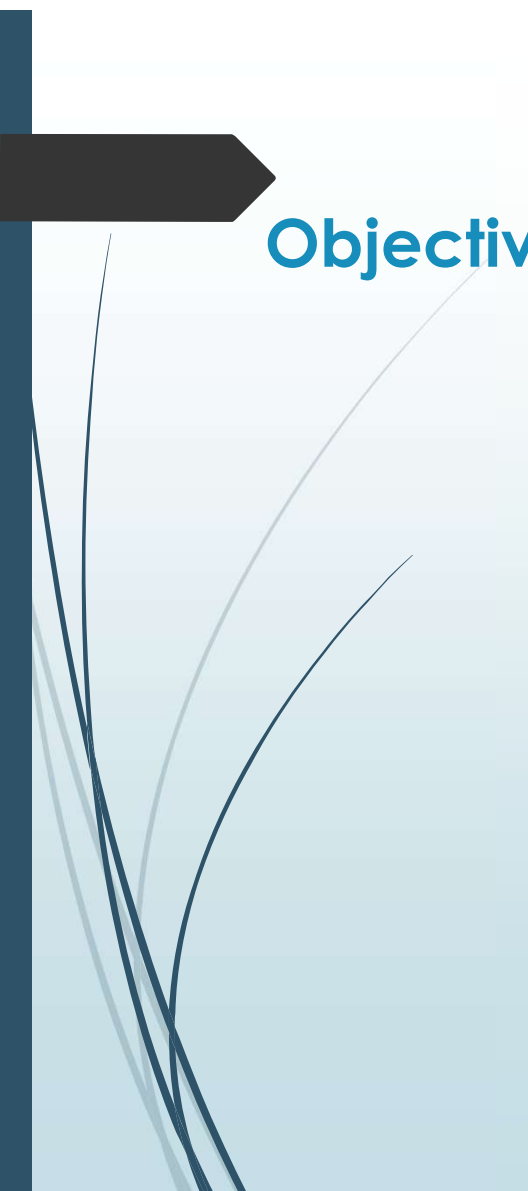
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This follows the uploaded reference presentation.



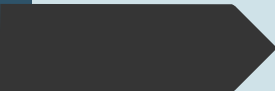
Project Overview

- Artificial Intelligence is transforming the fashion industry by enabling personalized shopping experiences through Virtual Try-On systems and AI recommendation engines.
- This internship focused on researching and understanding the technologies required to build an intelligent AI-powered fashion platform.
- **Major Research Areas**
 - Computer Vision
 - Body Segmentation
 - Pose Estimation
 - Body Measurement Extraction
 - Virtual Try-On
 - AI Recommendation Systems
 - 3D Avatar Generation



Objectives

- Study Artificial Intelligence applications in fashion.
- Understand Computer Vision fundamentals.
- Explore body segmentation and pose estimation.
- Research Virtual Try-On architectures.
- Analyse fashion datasets.
- Study garment warping techniques.
- Explore AI recommendation systems.
- Design a complete AI workflow.



Technologies and Tools

Programming

Python

OpenCV

MediaPipe

TensorFlow

Development
Tools

Visual Studio
Code

GitHub

AI
Frameworks

Detectron2

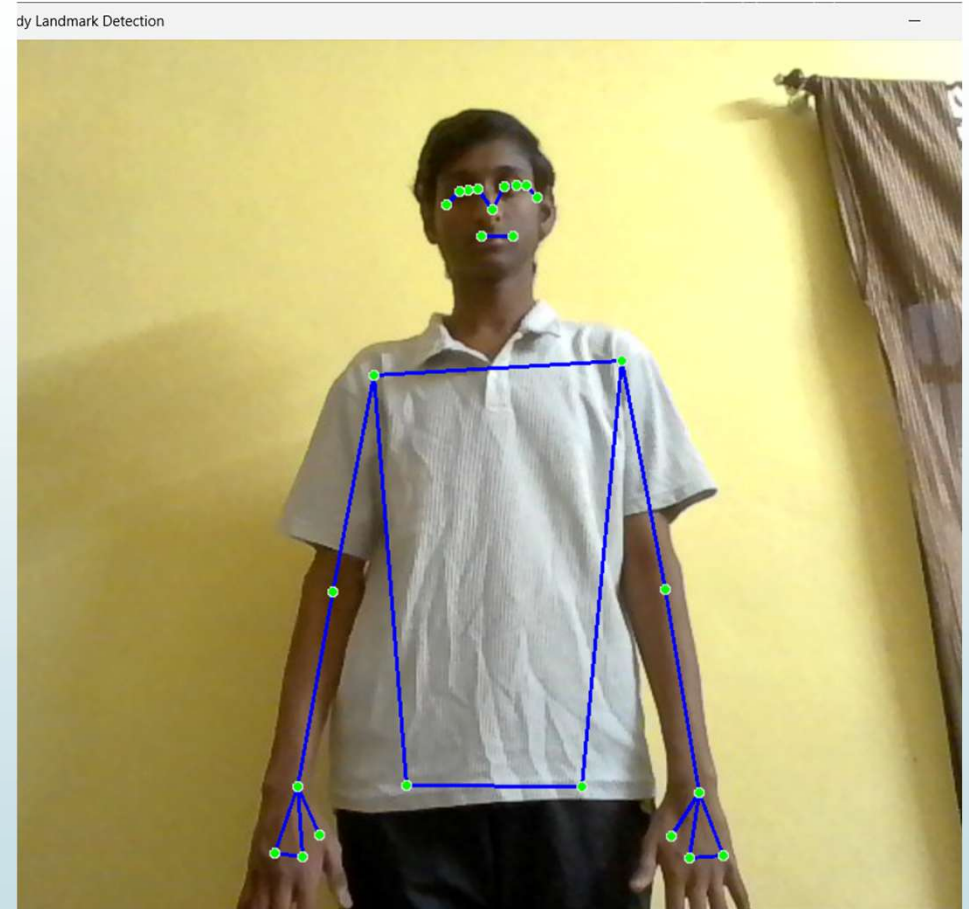
DeepLabV3

SCHP

OpenPose

Computer Vision in Fashion AI

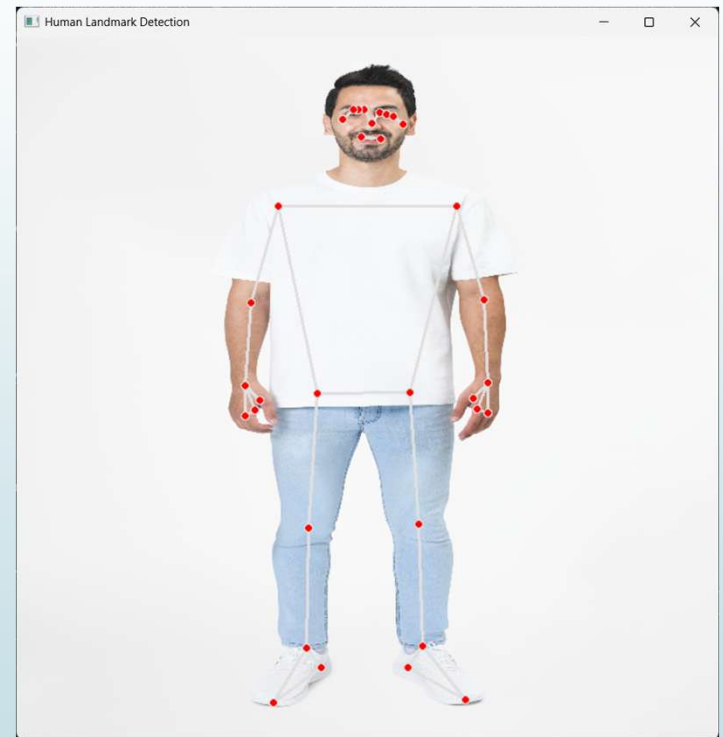
- Computer Vision enables machines to understand and analyze images.
- **Applications**
 - Human Detection
 - Body Segmentation
 - Pose Estimation
 - Landmark Detection
 - Background Removal
 - Image Processing



Pose Estimation

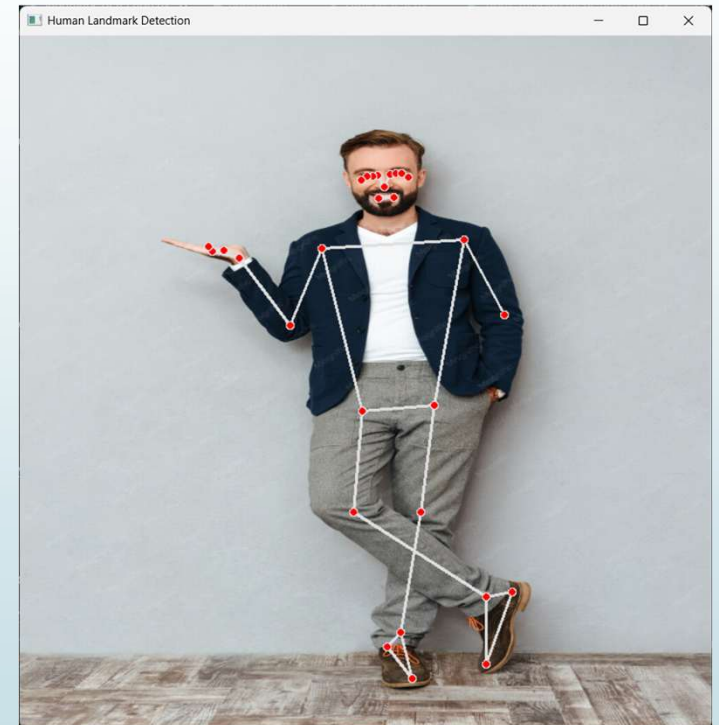
Human Pose Estimation

- Pose estimation identifies body joints and key points.
- **Models Studied**
 - MediaPipe Pose
 - OpenPose
- **Advantages**
 - Real-time processing
 - High accuracy
 - Landmark extraction



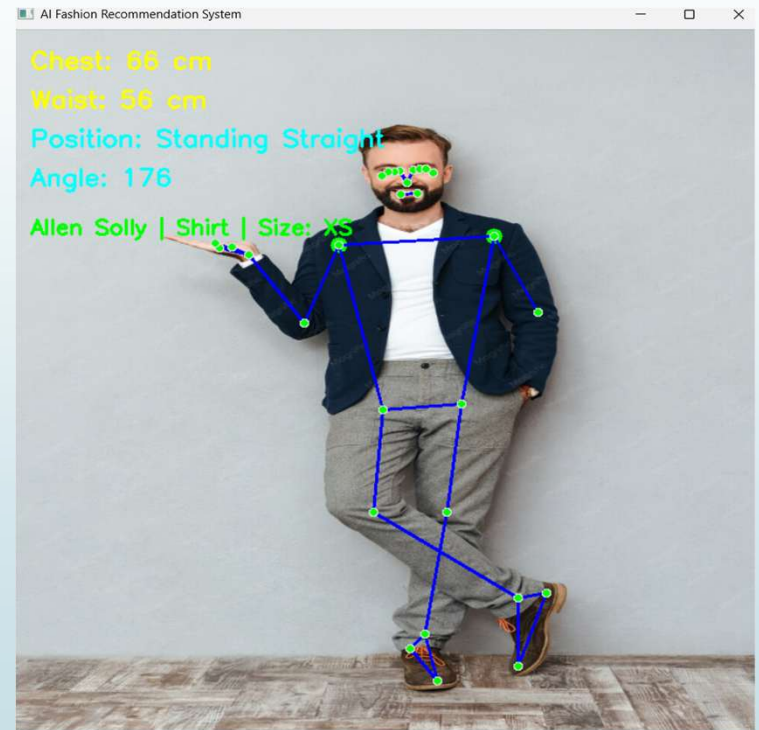
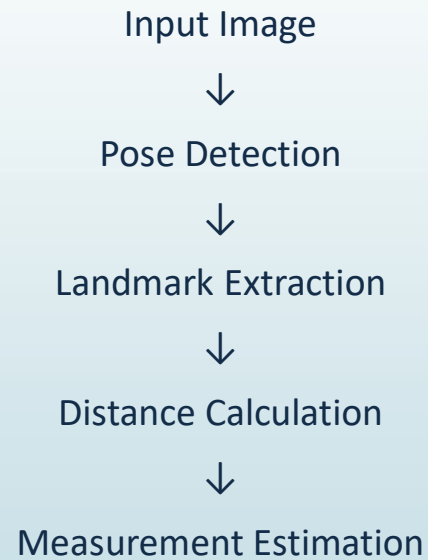
Body Landmark Detection

- Body landmarks are used to locate important body points.
- **Major Landmarks**
 - Shoulders
 - Elbows
 - Wrists
 - Hips
 - Knees
 - Ankles
- **Applications**
 - Body Measurement
 - Garment Alignment
 - Virtual Try-On



Body Measurement Extraction

Workflow





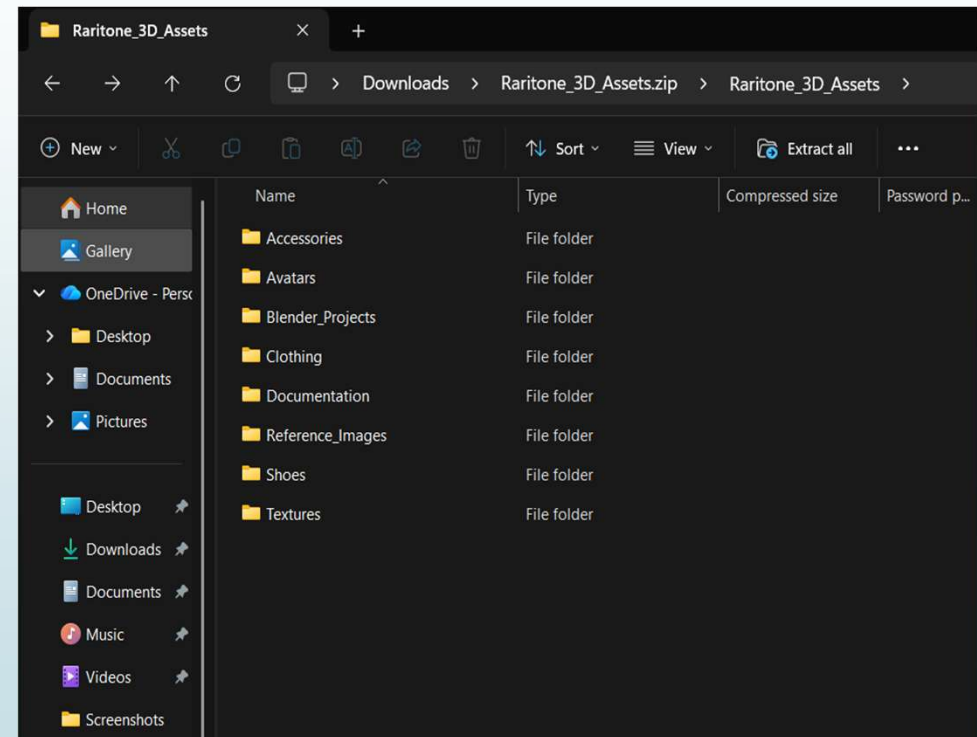
Fashion Datasets

Dataset	Purpose
DeepFashion	Fashion Recognition
DeepFashion2	Landmark Detection
VITON	Virtual Try-On
VITON-HD	High Resolution
Dress Code	Multi-category Clothing
Fashionpedia	Fashion Attributes

Dataset Preparation

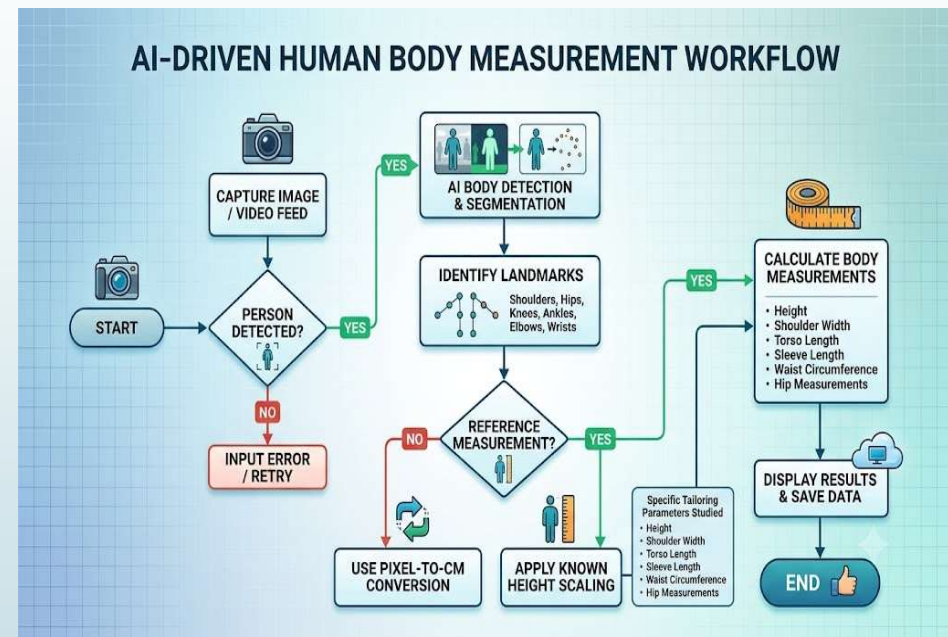
Dataset Preparation

- Data Collection
- Data Cleaning
- Metadata Generation
- Annotation
- Training / Validation Split
- Dataset Organization
- Dataset quality directly affects AI model performance.



AI-Based Virtual Try-On

- Virtual Try-On systems allow users to digitally wear garments before purchasing.
- Components**
 - Human Detection
 - Segmentation
 - Pose Estimation
 - Garment Processing
 - Image Generation



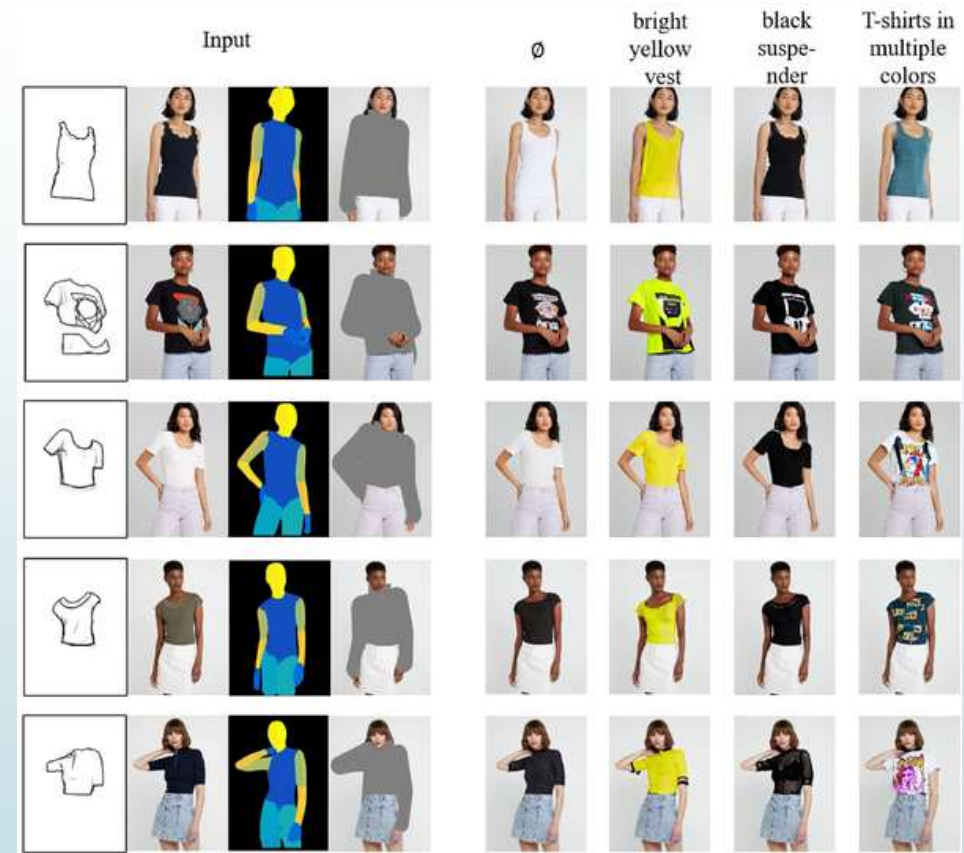


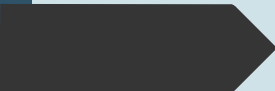
2D Virtual Try-On Models

Model	Strength
VITON	Basic Try-On
CP-VTON	Better Alignment
HR-VITON	High Resolution
StableVITON	Diffusion Model
IDM-VTON	Identity Preservation

Garment Warping

- **Garment Warping Techniques**
- **Thin Plate Spline (TPS)**
 - Smooth transformation
 - Landmark based
- **Geometric Matching Module (GMM)**
 - Deep Learning approach
 - Better garment fitting
- **Flow-Based Warping**
 - Texture preservation





AI-Based Cloth Blending

Techniques

- Alpha Blending
- Image Composition
- Edge Smoothing

Benefits

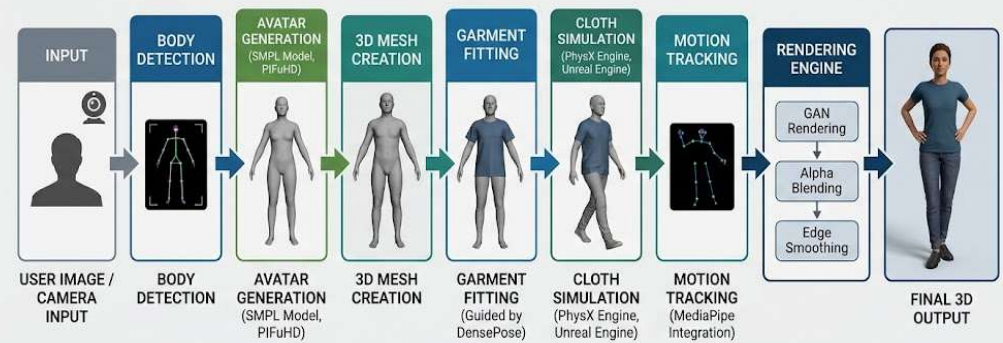
- Natural appearance
- Better garment integration
- Improved realism

3D Avatar Generation

Models Studied

- SMPL
- PIFuHD
- ICON
- HumanNeRF
- 3D Gaussian Splatting

3D AVATAR GENERATION WORKFLOW



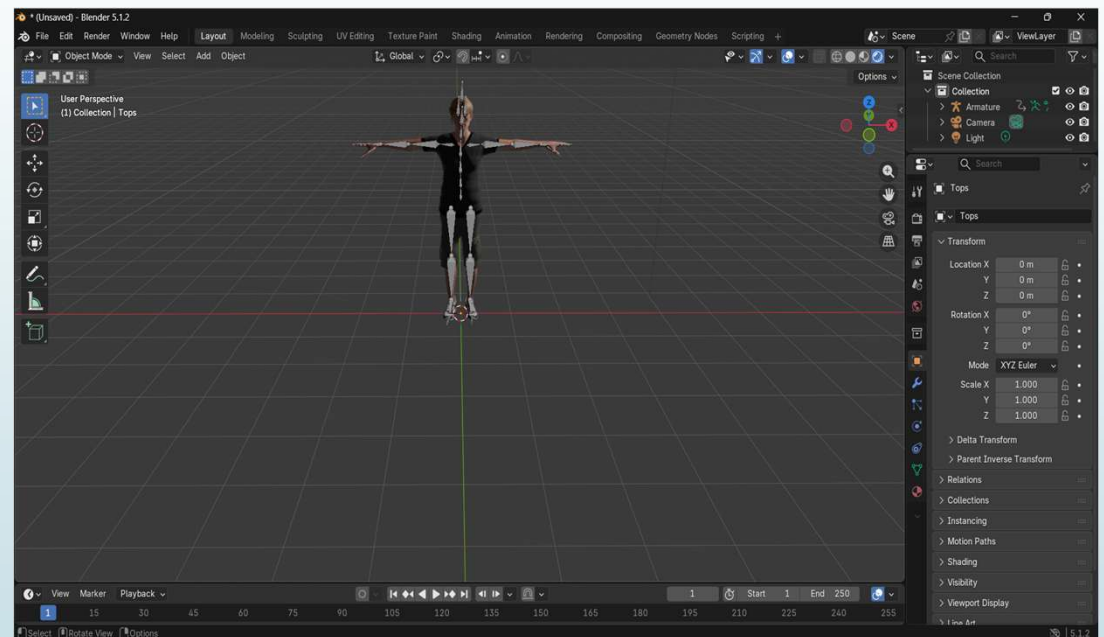
Cloth Simulation

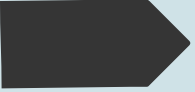
► Cloth Simulation Technologies

- Unity Cloth
- NVIDIA PhysX
- Marvelous Designer
- Blender Cloth Simulation

► Applications

- 3D Fashion
- Avatar Animation
- Digital Clothing



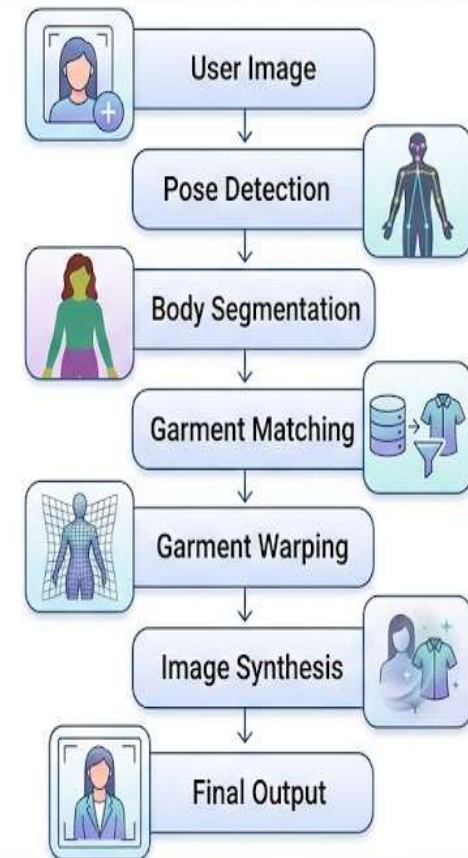


AI Recommendation Systems

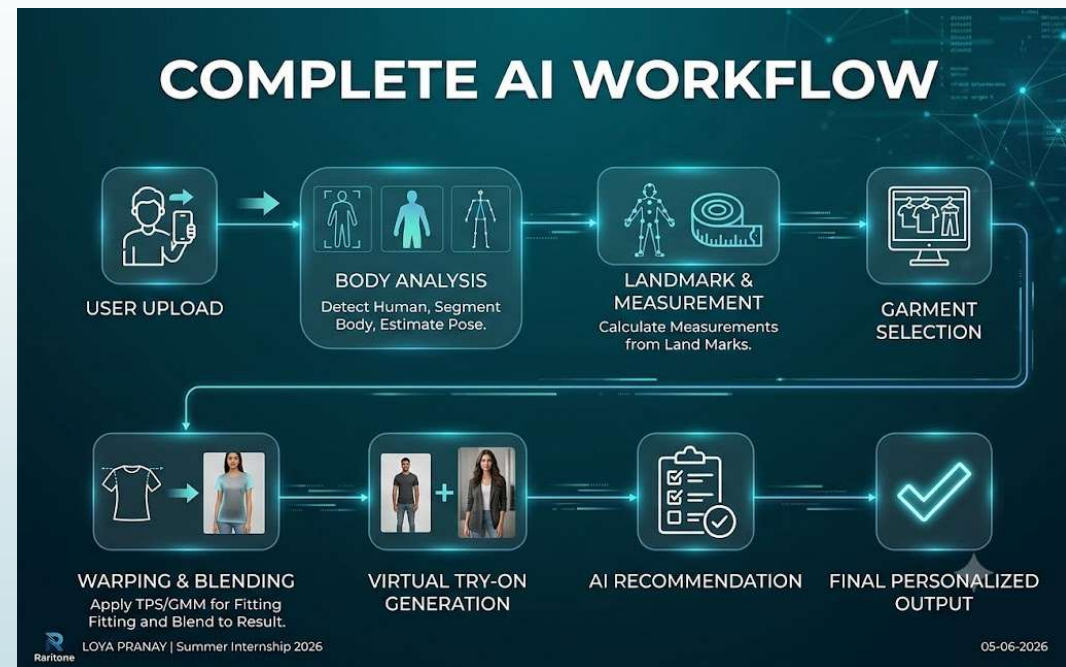
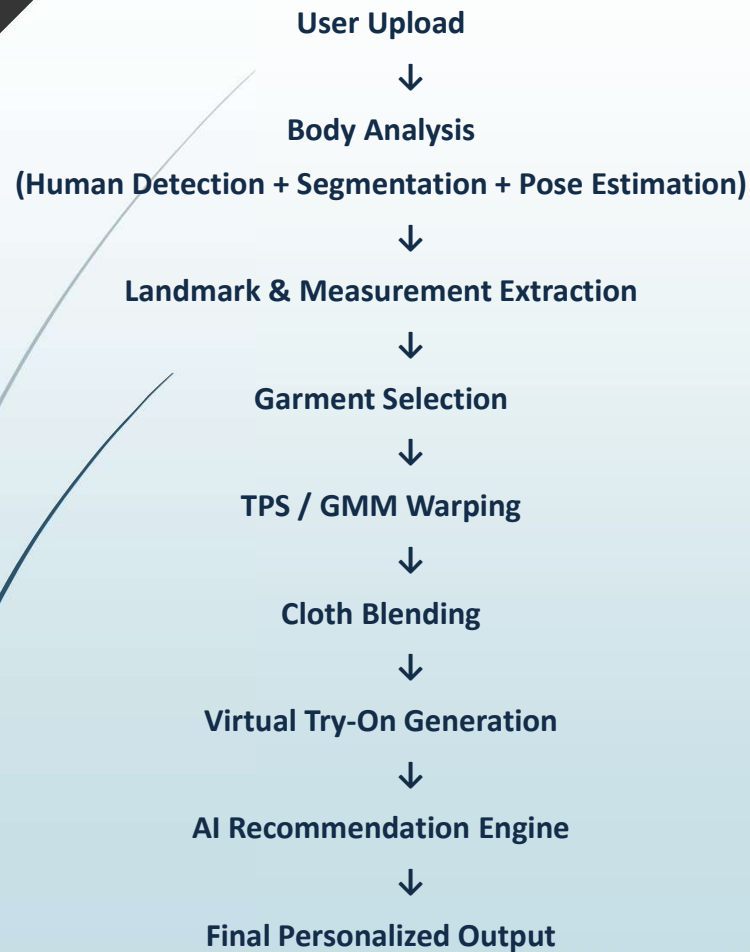
- Content-Based Filtering
- Collaborative Filtering
- Hybrid Recommendation
- Deep Learning Recommendation
- Purpose:
- Generate personalized outfit suggestions.

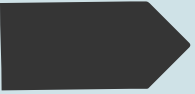
AI Recommendation Workflow

■ Recommendation Pipeline



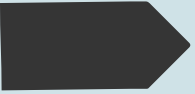
Complete AI Workflow





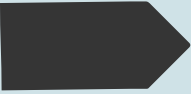
Challenges Faced

- Understanding Virtual Try-On systems.
- Learning advanced AI models.
- Dataset organization.
- Garment warping concepts.
- 3D avatar generation.
- Hardware limitations.
- Managing multiple research topics.



Solutions Applied

- Independent learning.
- Research paper analysis.
- Comparative studies.
- Workflow-based understanding.
- Structured documentation.
- GitHub organization.
- Modular learning approach.



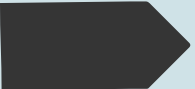
Skills Developed

► Technical Skills

- Artificial Intelligence
- Computer Vision
- Machine Learning
- OpenCV
- MediaPipe
- GitHub
- Technical Documentation

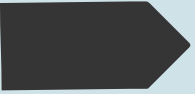
► Professional Skills

- Research
- Problem Solving
- Analytical Thinking
- Independent Learning



Key Learnings

- AI-powered Virtual Try-On workflow.
- Body segmentation and landmark detection.
- Body measurement extraction.
- Fashion datasets and annotations.
- Garment warping techniques.
- AI recommendation systems.
- 3D avatar generation.



Future Improvements

- Real-time Virtual Try-On.
- Mobile application deployment.
- Full 3D avatar support.
- Diffusion-based AI models.
- Personalized recommendation engines.
- Multi-angle garment visualization.
- Large-scale fashion datasets.



Conclusion

- The AI/ML Internship provided valuable exposure to Artificial Intelligence, Computer Vision, and modern fashion technologies.
- The project explored body segmentation, pose estimation, body measurement extraction, Virtual Try-On systems, garment warping, recommendation engines, and 3D avatar generation.
- The internship strengthened technical knowledge, research methodology, documentation skills, and problem-solving abilities while providing a strong foundation for future work in AI-powered fashion and intelligent digital systems.



Thank You

I would like to express my sincere gratitude to **Raritone** for providing me with the opportunity to be a part of the **Summer Internship 2026** as an **AI/ML Intern**.

I am thankful to the entire team and mentors for their guidance, support, and encouragement throughout this journey.

Thank You for Giving Me This Wonderful Opportunity.

LOYA PRANAY